

QUARTS for Probabilistic Traffic Forecasting: A Controlled Evaluation of Quantum and Classical Residual Calibration

Abstract

Quantum machine learning has been proposed for traffic forecasting, yet evidence from network traffic settings remains limited. We present QUARTS, a Quantum Uncertainty Aware Residual Traffic System, to test whether an entangled variational quantum circuit improves point forecasts or predictive intervals under identical experimental conditions. Experiments used METR-LA, PEMS-BAY, PeMSD4, and PeMSD8 with matched separable quantum, Fourier, multilayer perceptron, and unchanged backbone controls. Across five confirmatory seeds, entangled QUARTS obtained mean absolute errors of 3.897 on METR-LA and 2.030 on PEMS-BAY, which were 32.99% and 31.84% above STAEformer. Differences from the separable quantum and Fourier heads remained below 0.40%, providing no evidence that entanglement produced a predictive gain. A classical Fourier regime aware calibrator reduced weighted interval score by 14.58% relative to constant intervals and by 0.68% relative to a multilayer perceptron on prospective PeMSD4 test data, with 89.66% coverage. This advantage reversed for one of three independently trained backbones, and mean coverage declined to 88.78%. A conformal quantile variant failed its PeMSD8 validation gate, so the test split remained untouched. The results show that strong forecasting backbones, matched nonlinear controls, and explicit validation gates are necessary for credible assessment of quantum traffic models.

Keywords: traffic forecasting, quantum machine learning, variational quantum circuits, probabilistic forecasting, uncertainty calibration, conformal prediction

1 Introduction

Traffic forecasts support network management, traveller information, congestion mitigation, and operational planning. Their practical value depends on two related properties. The central estimate should remain accurate over several future horizons, and the accompanying uncertainty should indicate when that estimate is less dependable. Both properties are difficult to obtain because traffic observations are coupled through a road network, change with recurring temporal regimes, and respond to incidents, missing measurements, and other disturbances. Reviews of the field therefore

identify spatial dependence, temporal nonstationarity, transfer between networks, and trustworthy uncertainty as continuing problems rather than settled implementation details [1, 2].

Classical traffic forecasting has advanced rapidly on public sensor benchmarks. Diffusion convolution introduced directed graph propagation into recurrent forecasting [3]. Spatial and temporal graph convolution, adaptive adjacency learning, attention, and node specific embeddings subsequently improved the representation of network structure and periodic behaviour [4–8]. More recent work shows that much of the predictive gain can arise from a suitable representation rather than an increasingly elaborate forecasting block. STID uses spatial and temporal identity information as a compact baseline [9], while STAEformer combines adaptive spatial and temporal embeddings with a standard Transformer and reports strong results on common benchmarks [10]. These developments make a weak classical comparator unsuitable for assessing a quantum model.

Probabilistic forecasting adds a second requirement. A model can obtain a reasonable mean absolute error while producing intervals that are too narrow, too wide, or unstable under a change of traffic regime. Proper scoring rules provide a principled way to evaluate predictive distributions [11]. The weighted interval score combines interval sharpness with penalties for observations outside the reported bounds and is suitable when several central intervals are available [12]. Conformal prediction offers a complementary route to empirical coverage without requiring a fully specified conditional distribution [13]. Conformalized quantile regression can adapt interval width to heterogeneous residual variance [14], although sequential traffic observations do not automatically satisfy the exchangeability assumptions used by standard conformal arguments. Methods developed for dynamic time series illustrate both the promise and the care needed in this setting [15].

Quantum machine learning provides trainable feature maps and variational circuits that can be coupled to classical networks [16, 17]. Data reuploading repeatedly encodes classical features and can generate rich nonlinear functions with a small number of qubits [18]. The resulting function class has a Fourier interpretation that depends on the encoding and observable [19]. This interpretation creates a demanding attribution problem. If a variational quantum circuit improves a forecast, the comparison must establish whether the gain comes from entanglement, from a periodic nonlinear basis that a classical Fourier layer can reproduce, or simply from adding trainable parameters. Broad quantum machine learning benchmarks have found that classical methods often remain stronger and that removing entanglement can leave performance unchanged or improve it on small learning tasks [20].

Transportation research has begun to examine quantum optimization, sensing, and prediction [21, 22]. Direct traffic forecasting evidence is still limited. Schetakakis et al. studied data reuploading quantum neural networks on a single urban detector and identified multiple input, multiple output, and transfer settings as important extensions [23]. Rajagopal et al. reported a quantum inspired graph formulation for traffic flow prediction [24]. These studies motivate further investigation, but they do not resolve whether an executable entangled circuit contributes beyond matched classical nonlinear controls on standard network wide forecasting tasks.

This study addresses that question through QUARTS, the Quantum Uncertainty Aware Residual Traffic System. A classical forecasting backbone produces a base forecast and local latent state. A shared calibration head then estimates a point correction, an uncertainty scale, or interval radii. The head can be an entangled variational quantum circuit, a circuit with the entanglers removed, a classical Fourier module, a multilayer perceptron, or no learned calibrator. Every learned head receives the same local information and is evaluated under a common split, optimization budget, and metric definition. Later variants freeze STAEformer so that calibration cannot be mistaken for a stronger forecasting backbone.

The investigation is organized around four questions. First, does the entangled head improve point forecasting over the unchanged backbone and STAEformer? Second, does it outperform separable quantum, Fourier, and multilayer perceptron controls? Third, can regime aware residual calibration improve interval quality while maintaining nominal coverage? Fourth, do observed gains survive independent backbone initialization and a new dataset? These questions are evaluated with five confirmatory seeds on METR-LA and PEMS-BAY, prospective testing on PeMSD4, and a validation only transfer study on PeMSD8.

The study makes four contributions. First, it provides a controlled architecture that isolates the function of a small shared quantum calibrator from the network scale forecasting backbone. Second, it evaluates entanglement against a separable circuit and a Fourier control that directly reflects the known spectral structure of data reuploading models. Third, it links point accuracy, interval quality, calibration, computation time, circuit sensitivity, missing input robustness, and paired statistical analysis in one traceable protocol. Fourth, it reports the complete sequence of favourable, negative, and inconclusive findings. The confirmatory study does not support a quantum advantage. A classical Fourier regime calibrator shows a small prospective interval score improvement, but that improvement changes sign for one independently trained backbone and does not satisfy the coverage requirement in aggregate. The resulting contribution is therefore an evidence bounded account of what residual calibration achieved, where it failed, and why entanglement cannot be credited for the observed behaviour.

2 Related work

2.1 Network traffic forecasting

Traffic forecasting methods have moved from recurrent models with fixed spatial structure toward architectures that learn interactions jointly from data. DCRNN represents directed traffic propagation as diffusion on a graph [3], while STGCN uses graph and temporal convolutions without recurrent units [4]. Graph WaveNet and AGCRN learn adaptive relations that can supplement or replace a prescribed road graph [5, 6]. ASTGCN and GMAN use attention to distinguish relevant locations and times [7, 8]. MTGNN extends graph learning to general multivariate time series [25]. PDFormer introduces propagation delay awareness, which is particularly relevant when congestion reaches nearby sensors at different times [26].

Recent results also question whether architectural complexity alone explains progress. STID obtains competitive accuracy through spatial and temporal identity embeddings [9]. STAEformer similarly uses adaptive embeddings to supply a Transformer with traffic specific context [10]. QUARTS therefore treats a strong backbone as a necessary comparator. The study does not claim that a small quantum head can replace graph scale representation learning. It asks the narrower question of whether such a head adds useful local residual structure after the backbone has formed its representation.

2.2 Probabilistic forecasting and calibration

Probabilistic forecasting evaluates a distribution or a set of intervals rather than a single trajectory. DeepAR demonstrated how a shared autoregressive model can learn probabilistic forecasts across many related series [27]. Proper scoring rules discourage a forecaster from improving apparent coverage merely by widening every interval [11]. The weighted interval score is especially useful when evaluation data contain a finite set of central intervals [12].

Conformal prediction separates the fitted predictor from a calibration step and offers finite sample marginal coverage under exchangeability [13]. Conformalized quantile regression retains feature dependent interval width and then corrects residual miscoverage [14]. Time ordered traffic data complicate the usual guarantee because adjacent observations are dependent and their distribution may drift. EnbPI addresses related difficulties with sequential residual updates [15]. In this study, conformal calibration is used as an empirical procedure on chronological validation data. Coverage is reported, and no claim of distribution free conditional coverage is made.

2.3 Quantum learning and matched controls

Variational quantum models commonly combine a classical input map, trainable quantum gates, measurement, and a classical readout [16, 17]. Quantum feature maps can represent nonlinear similarities in a Hilbert space [28], and data reuploading increases the expressiveness of small circuits [18]. Their representational power can exceed that of some similarly sized classical networks in selected settings [29]. However, expressibility is not equivalent to useful generalization. Highly expressive circuits may be difficult to train [30]; gradients can vanish with circuit size or unsuitable initialization [31]; and noise can further suppress gradients [32].

The Fourier representation of encoded quantum models is central to the present controls [19]. A periodic classical layer can reproduce an important part of the inductive bias that might otherwise be attributed to the circuit. The separable circuit provides a second control by retaining encoding, trainable rotations, measurement, and readout while removing controlled gates. These controls implement the recommendation emerging from broad benchmarks that quantum models should be judged against strong, structurally matched classical alternatives [20].

2.4 Research gap

Existing traffic studies establish that quantum and quantum inspired prediction is technically possible [23, 24], while transportation reviews identify wider opportunities in optimization and sensing [21, 22]. Three gaps remain. Network wide evaluation is scarce, the contribution of entanglement is rarely isolated from periodic nonlinear processing, and probabilistic performance is often omitted. QUARTS addresses these gaps through four public networks, identical inputs for all calibration heads, explicit promotion gates, and a record that includes failed transfer and robustness tests.

3 Problem formulation and study design

3.1 Forecasting task

Let $G = (V, E, A)$ denote a sensor network with $N = |V|$ nodes and weighted adjacency matrix A . At time t , the observed traffic state is $X_t \in \mathbb{R}^{N \times F}$. Given twelve historical observations, the task is to estimate the next twelve observations:

$$f_\theta : (X_{t-11}, \dots, X_t, A, C_t) \mapsto (\hat{Y}_{t+1}, \dots, \hat{Y}_{t+12}), \quad (1)$$

where C_t contains available temporal covariates. Twelve steps correspond to one hour for the five minute datasets used here. Missing or invalid targets are excluded through a binary mask M .

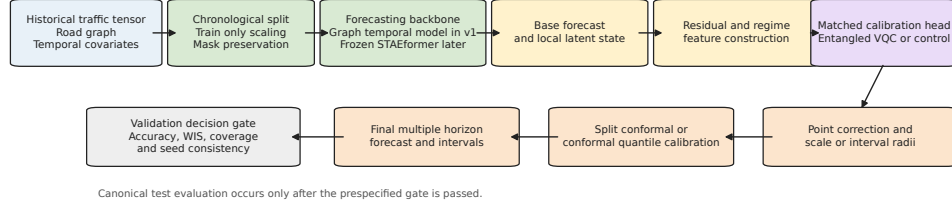
For probabilistic evaluation, the forecaster also returns central intervals $[L_{t+h,n}^{(\alpha)}, U_{t+h,n}^{(\alpha)}]$ for $\alpha \in \{0.1, 0.2, 0.5\}$. The primary interval is the nominal 90% interval obtained with $\alpha = 0.1$. A useful interval method must balance coverage and sharpness. Coverage alone is insufficient because an uninformatively wide interval can cover every observation.

3.2 Research questions and decision logic

The experiments answer four prespecified questions. RQ1 asks whether the entangled residual head improves point accuracy relative to its unchanged backbone and STAE-former. RQ2 asks whether it improves over the separable circuit, Fourier layer, and multilayer perceptron. RQ3 asks whether conditional residual calibration lowers WIS while preserving acceptable 90% coverage. RQ4 asks whether a selected improvement remains directionally consistent across independent backbone seeds and transfers to PeMSD8.

Each extension was evaluated behind a validation gate. A branch advanced only when it improved the declared primary metric, remained within the declared coverage range when intervals were involved, and satisfied any seed consistency requirement. A failed branch was retained as evidence but was not promoted. In particular, the PeMSD8 test split was not opened after the conformal quantile branch failed validation. This separation between model development and final testing limits the opportunity to select a favourable test outcome.

(a) End to end forecasting and calibration workflow



(b) Parameter matched head comparison and entangled circuit

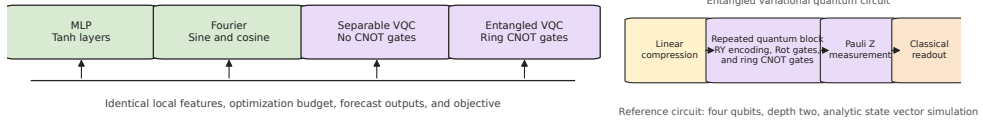


Fig. 1 QUARTS system architecture. The upper panel shows the chronological data, forecasting, residual calibration, conformal adjustment, evaluation, and decision stages. The lower panel isolates the four matched calibration heads and the internal structure of the variational quantum circuit.

4 QUARTS system architecture and method

4.1 System overview

Figure 1 presents the complete system. A chronological preprocessing stage constructs masked input windows, scales values with training data only, and supplies the network graph and time covariates. The forecasting backbone produces a base trajectory B and a latent node representation H . A shared calibration head operates on the same representation for every treatment. Depending on the experiment, it estimates an additive point correction, a positive uncertainty scale, or the mixture weights of ordered interval experts. Split conformal or conformal quantile calibration uses only the assigned validation segment. A decision gate then determines whether the branch may progress to a protected test split.

4.2 Forecasting backbones

The reference QUARTS backbone combines graph propagation with a gated recurrent encoder. For an input X , graph propagated features AX are concatenated with the original features and encoded for every node. A linear horizon head maps the final hidden state $H \in \mathbb{R}^{N \times d}$ to the base forecast $B \in \mathbb{R}^{12 \times N}$. The unchanged version of this backbone is retained as a control.

The later uncertainty studies use a frozen STAEformer backbone. Freezing is methodologically important because all interval heads then receive identical point forecasts and representations. Any difference in WIS can be attributed to residual

calibration rather than to a change in point forecasting capacity. STAEformer is also trained independently with seeds 42, 73, and 202 for the PeMSD4 replication.

4.3 Entangled and separable quantum heads

The reference quantum head uses four qubits and depth two. A linear layer compresses each node representation to q values. The bounded angles are

$$z = \pi \tanh(W_c H + b_c). \quad (2)$$

At each circuit layer, $R_Y(z_j)$ reuploads the corresponding feature on qubit j , followed by a trainable $R_Z R_Y R_Z$ rotation. The entangled treatment applies a ring of controlled NOT gates after these local rotations. The separable treatment removes this ring and changes nothing else. Pauli Z expectation values form the vector m , and a linear readout produces the required horizons:

$$(\Delta, \tilde{s}) = W_o m + b_o, \quad s = \text{softplus}(\tilde{s}) + \epsilon. \quad (3)$$

For point forecasting, $\hat{Y} = B + \Delta$. For scale only experiments, the frozen location remains B . The main experiments use analytic state vector simulation without sampling shots. No claim about execution speed or accuracy on physical quantum hardware is made.

4.4 Classical matched heads

The Fourier control receives the same compressed node representation. At layer k , it forms

$$p_k = [\sin(kz + \phi_k), \cos(kz + \phi_k)], \quad z \leftarrow \tanh(W_k p_k + b_k). \quad (4)$$

It therefore exposes periodic functions related to the spectral structure of data reuploading while remaining fully classical. Width was selected to keep its trainable parameter count close to that of the quantum head. The multilayer perceptron uses a compact hyperbolic tangent hidden layer. The unchanged backbone supplies a no calibration control. Together, these treatments distinguish an entanglement effect from periodic nonlinearity, generic nonlinear capacity, and the effect of adding any calibration layer.

4.5 Regime aware interval calibration

The regime aware calibrator contains $K = 3$ shared interval experts. A head maps the node representation to logits, which become mixture weights

$$\pi_{n,k} = \frac{\exp(g_k(H_n))}{\sum_{j=1}^K \exp(g_j(H_n))}. \quad (5)$$

Positive increments are summed in reverse order to obtain nested radii for the three interval levels. The mixture of expert radii produces $r^{(\alpha)}$, and the symmetric interval

is $B \pm r^{(\alpha)}$. The asymmetric extension learns separate lower and upper increments. The ordered construction prevents a nominally wider interval from being contained inside a narrower one.

After training, horizon and level specific conformal factors are estimated from a dedicated chronological calibration segment. SA-CQR instead trains lower and upper residual quantiles with pinball loss and applies a finite sample conformal adjustment. These procedures calibrate the frozen forecast rather than altering its location.

4.6 Objectives and evaluation measures

Point performance is reported with mean absolute error, root mean squared error, and weighted absolute percentage error. For observed targets $\mathcal{I} = \{i : M_i = 1\}$,

$$\text{MAE} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} |Y_i - \hat{Y}_i|. \quad (6)$$

For a central interval $[L_\alpha, U_\alpha]$, the interval score is

$$\text{IS}_\alpha = (U_\alpha - L_\alpha) + \frac{2}{\alpha}(L_\alpha - Y)\mathbf{1}\{Y < L_\alpha\} + \frac{2}{\alpha}(Y - U_\alpha)\mathbf{1}\{Y > U_\alpha\}. \quad (7)$$

The WIS used in training and evaluation combines absolute error at the location with interval scores at $\alpha \in \{0.1, 0.2, 0.5\}$, using the conventional $\alpha/2$ weights [12]. Lower WIS is better. The 90% empirical coverage and mean width are reported beside WIS so that score changes can be interpreted.

5 Experimental design

5.1 Datasets and preprocessing

Table 1 summarizes the four public traffic networks. METR-LA and PEMS-BAY contain speed measurements from 207 and 325 sensors. PeMSD4 and PeMSD8 contain traffic measurements from 307 and 170 detectors. All series use five minute sampling. METR-LA includes 34,272 time steps from March to June 2012, PEMS-BAY includes 52,116 time steps from January to June 2017, and PeMSD4 includes 16,992 time steps from January to February 2018. The dataset audit found no not a number values in METR-LA, although 8.109% of entries were zero and were masked according to the established preprocessing convention.

Every dataset was divided chronologically into 70% training, 10% validation, and 20% test partitions. Scaling statistics were calculated from the training partition only. Sliding windows did not cross partition boundaries. The validation partition was subdivided when early stopping, conformal calibration, and held out validation were all required. Test observations were never used to fit head parameters or conformal factors.

Table 1 Traffic datasets and their role in the study.

Dataset	Sensors	Steps	Input	Horizon	Role
METR-LA	207	34,272	12	12	Confirmatory and development
PEMS-BAY	325	52,116	12	12	Confirmatory
PeMSD4	307	16,992	12	12	Prospective replication
PeMSD8	170	17,856	12	12	Validation transfer

5.2 Comparators and implementation

The point study includes historical average and persistence references, the unchanged graph temporal backbone, MLP, Fourier, separable quantum, and entangled quantum residual heads, and the official STAEformer implementation. All QUARTS heads use the same hidden representation and twelve horizon outputs. The reference circuit has four qubits and two reuploading layers. The architecture study evaluates $q \in \{2, 4, 6\}$ and depths $d \in \{1, 2, 3\}$.

Models were trained for at most 30 epochs with Adam, learning rate 10^{-3} , weight decay 10^{-4} , batch size 256, early stopping patience seven, and minimum improvement 5×10^{-4} . Confirmatory point results use seeds 17, 42, 73, 101, and 202. The uncertainty branches use their declared head seeds, while the independent PeMSD4 study trains complete backbones with seeds 42, 73, and 202.

The experiments ran on Windows 11 with Python 3.12.13, PyTorch 2.13.0 built for CUDA 13.0, PennyLane 0.39, NumPy 2.0.2, pandas 2.2.3, SciPy 1.18.0, and scikit learn 1.9.0. The GPU was an NVIDIA GeForce RTX 5060 Laptop GPU with 8,151 MiB of memory. Quantum models were simulated on the same computer, which provides a controlled software comparison but not a hardware quantum benchmark.

5.3 Statistical analysis

Overall differences among point models were assessed separately for each dataset with the Friedman rank test [33]. Paired mean error differences were accompanied by percentile bootstrap 95% confidence intervals [34]. Holm adjustment controlled multiplicity for the paired comparisons. Cliff’s delta was also recorded as a distribution free effect size [35]. These paired analyses use the stored forecast errors and should not be interpreted as independent sensor samples.

For PeMSD4, Fourier and MLP WIS contributions were paired within each independently trained backbone. The decision did not rest on statistical significance alone. Directional consistency and the prespecified coverage range were required because a very large number of forecast instances can make a practically small difference statistically detectable.

5.4 Experiment gates

The study followed a staged sequence. Point forecasting first tested whether the entangled head warranted promotion. QUARTS-UQ then isolated uncertainty scale prediction. QUARTS-RAC tested regime selection on validation data. RAQ tested

Table 2 Confirmatory point forecasting over five seeds. Values are mean with standard deviation in parentheses. Lower values are better.

Dataset	Model	MAE	RMSE	MAPE (%)
METR-LA	Unchanged backbone	3.8763 (0.0014)	7.7322 (0.0097)	10.9696 (0.0437)
	MLP head	3.9003 (0.0079)	7.8123 (0.0200)	11.1479 (0.0760)
	Fourier head	3.9039 (0.0035)	7.8275 (0.0172)	11.1575 (0.0456)
	Separable VQC	3.8923 (0.0064)	7.7948 (0.0163)	11.0770 (0.0483)
	Entangled VQC	3.8969 (0.0054)	7.8023 (0.0122)	11.1382 (0.0539)
	STAEformer	2.9302 (0.0077)	5.9603 (0.0144)	8.0754 (0.0185)
PEMS-BAY	Unchanged backbone	1.9909 (0.0023)	4.6805 (0.0161)	4.5131 (0.0189)
	MLP head	2.0305 (0.0059)	4.8128 (0.0140)	4.6246 (0.0269)
	Fourier head	2.0376 (0.0052)	4.8505 (0.0240)	4.5999 (0.0216)
	Separable VQC	2.0248 (0.0036)	4.8021 (0.0067)	4.5813 (0.0278)
	Entangled VQC	2.0295 (0.0043)	4.8222 (0.0290)	4.5735 (0.0194)
	STAEformer	1.5394 (0.0023)	3.5527 (0.0112)	3.4520 (0.0293)

asymmetric intervals. A favourable Fourier result was next evaluated prospectively on PeMSD4 and repeated with two additional independently trained backbones. Recent conformal repair and SA-CQR transfer were conducted only after the corresponding failure was observed. Table 3 in Section 6 reports every decision, including failed branches.

6 Results

6.1 Confirmatory point forecasting

Table 2 and Fig. 2 show the five seed results. Entangled QUARTS obtained MAE 3.8969 on METR-LA and 2.0295 on PEMS-BAY. The unchanged graph temporal backbone was better on both datasets. STAEformer obtained 2.9302 and 1.5394, making the entangled head 32.99% and 31.84% worse. The result rejects RQ1 for the tested architecture.

The matched comparisons answer RQ2. Entangled QUARTS was 0.120% worse than the separable circuit on METR-LA and 0.233% worse on PEMS-BAY. It was 0.178% and 0.397% better than Fourier. These differences are small relative to the gap from STAEformer and do not establish an entanglement benefit. The paired entangled minus separable MAE differences were 0.0055, with 95% confidence interval [0.0036, 0.0076], and 0.0047, with interval [0.0038, 0.0055], respectively. Cliff’s delta remained at or below 0.0022. The entangled minus Fourier differences were -0.0073 , with interval $[-0.0096, -0.0053]$, and -0.0078 , with interval $[-0.0100, -0.0055]$, with absolute Cliff’s delta below 0.0033. The Friedman tests rejected equality among all treatments on METR-LA ($\chi^2_F = 21.23$, $p = 0.00073$) and PEMS-BAY ($\chi^2_F = 20.43$, $p = 0.00104$), a result driven primarily by the much larger difference from STAEformer.

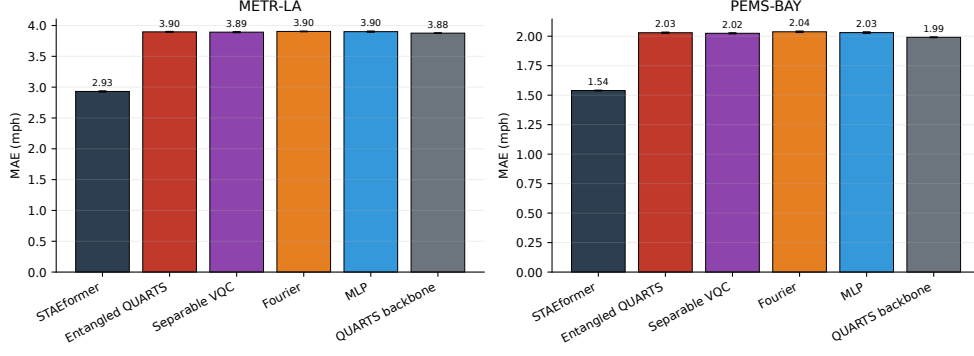


Fig. 2 Confirmatory MAE and RMSE on METR-LA and PEMS-BAY. Error bars summarize variation across five seeds.

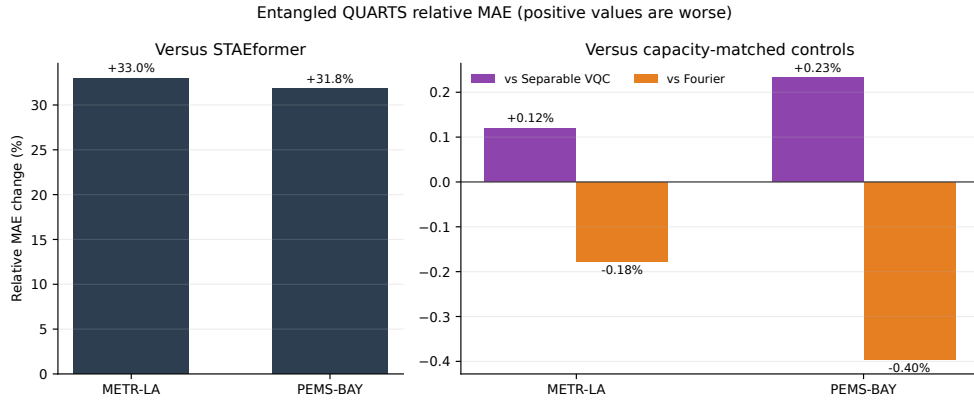


Fig. 3 Relative effects of the entangled quantum head against matched controls. Values near zero indicate that the compared heads behave similarly.

6.2 Architecture and efficiency

Increasing circuit size did not produce a monotonic accuracy gain. Across nine METR-LA configurations, MAE ranged from 3.8839 to 3.9046. The best value occurred with six qubits and one layer, while six qubits and deeper circuits were slightly worse. Inference time increased from 1.069 s for two qubits and one layer to 16.760 s for six qubits and three layers. The reference four qubit, depth two quantum head contained 4,140 trainable parameters, close to the Fourier head’s 4,196 parameters. Nevertheless, its METR-LA total training time was 1,544.3 s compared with 162.2 s for Fourier. Figure 4 places these costs beside accuracy. The simulator therefore offered no computational advantage.

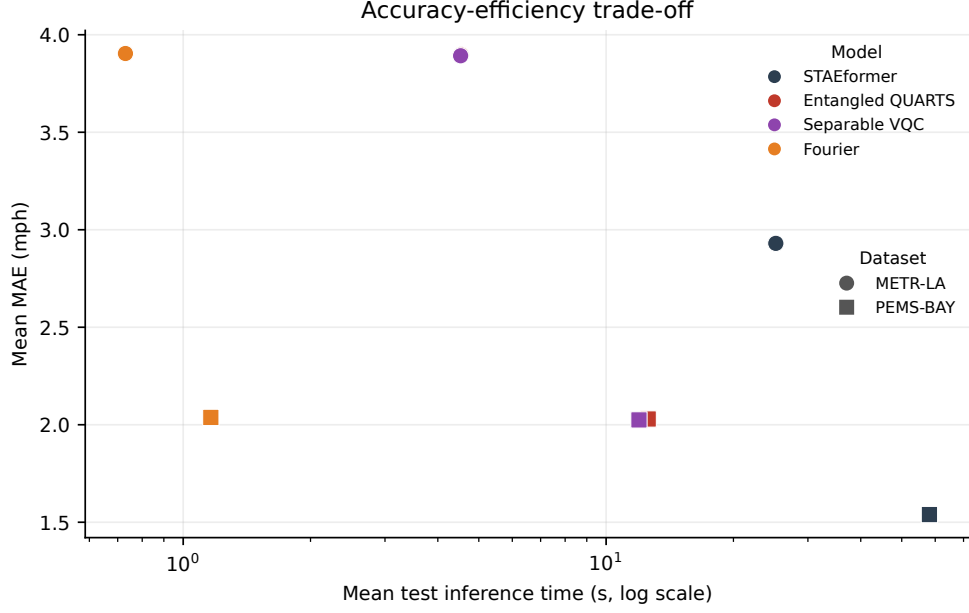


Fig. 4 Accuracy and measured computational cost for the confirmatory models. Times refer to the reported local GPU and simulated circuit environment.

6.3 Uncertainty and regime calibration

Figure 5 consolidates the uncertainty stages. On METR-LA, QUARTS-UQ produced WIS 2.0045 and 89.89% coverage. The separable circuit achieved a slightly lower WIS of 1.9999, so the entangled proposal was 0.23% worse and failed its required 5% improvement gate. In the validation only regime study, Fourier-RAC obtained the best WIS, 1.6869, with 90.65% coverage. Entangled RAC obtained 1.6986, 0.70% worse than Fourier. The asymmetric Fourier variant increased WIS from 1.6869 to 1.7267 and raised coverage to 92.17%. It therefore failed both the score and coverage criteria.

These findings answer RQ3 for METR-LA. Conditional calibration improved over a constant interval model, but the improvement was classical rather than quantum. Symmetric radii were more reliable than the tested asymmetric extension. A gated point correction initialized near the unchanged STAEformer prediction also produced MAE values between 2.9276 and 2.9277 for all learned heads, so it did not create a meaningful point forecasting gain.

6.4 Prospective PeMSD4 evaluation

The Fourier regime calibrator was selected before opening the PeMSD4 test split. With backbone seed 42, its WIS was 26.8820, compared with 31.4672 for constant intervals and 27.0663 for MLP-RAC. These values correspond to improvements of 14.58% and

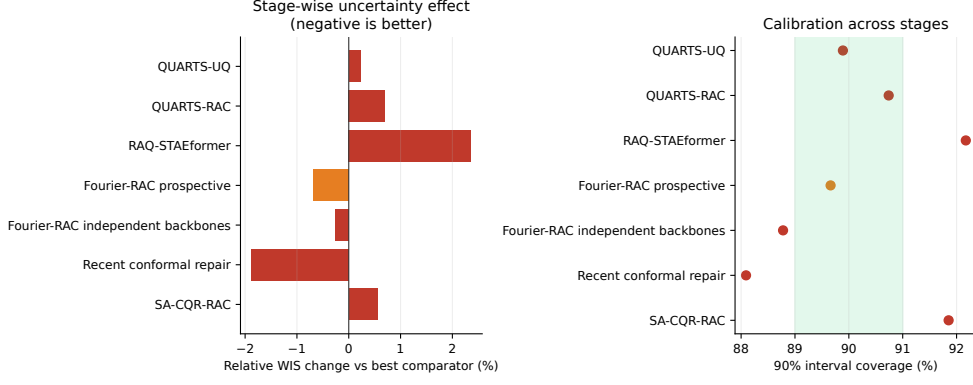


Fig. 5 Synthesis of the uncertainty experiments. WIS is evaluated together with 90% empirical coverage and the decision status of each stage.

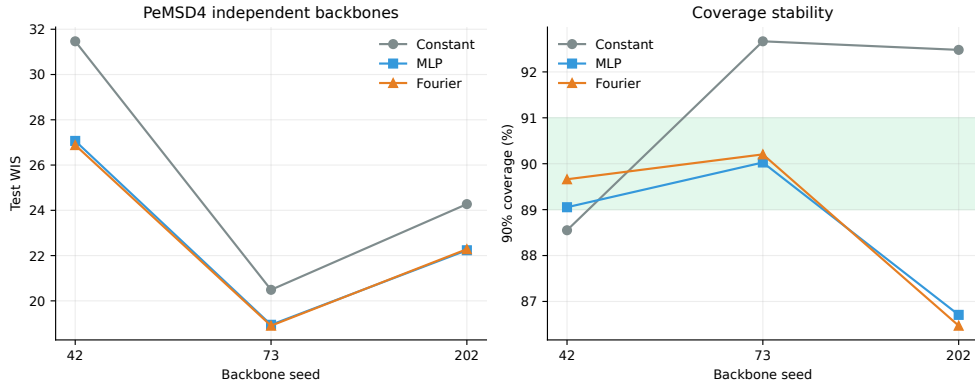


Fig. 6 PeMSD4 WIS and 90% coverage for three independently trained STAEformer backbones. The Fourier advantage reverses for seed 202, and its mean coverage falls below the gate.

0.68%, and Fourier coverage was 89.66%. This was the strongest prospective result, but its small margin over MLP required independent replication.

Across seeds 42, 73, and 202, mean Fourier WIS was 22.6892, compared with 22.7480 for MLP. Mean Fourier coverage was 88.78%, below the declared lower bound of 89%. More importantly, Fourier minus MLP WIS changed from -0.1843 and -0.0402 for seeds 42 and 73 to $+0.0479$ for seed 202. All paired confidence intervals excluded zero because each run contained more than one million forecast instances, but the sign reversal and aggregate undercoverage failed the replication gate. Figure 6 shows both the average and the seed level behaviour.

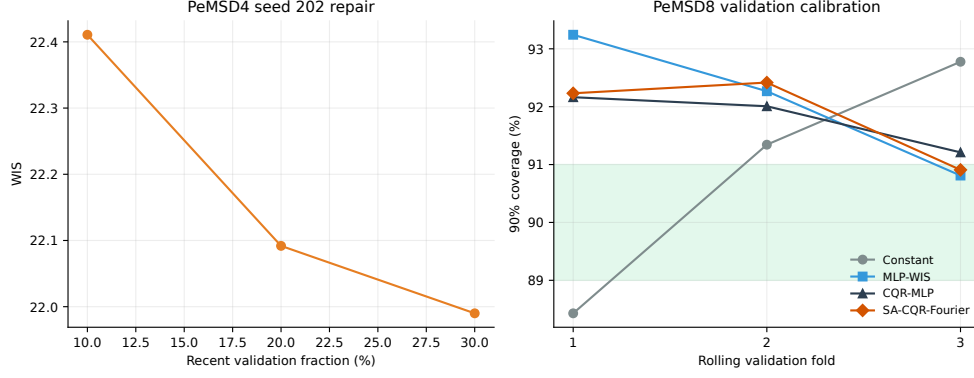


Fig. 7 Recent conformal repair on PeMSD4 and validation only SA-CQR transfer to PeMSD8. Neither branch satisfied its complete decision rule.

Table 3 Experiment gate summary. Relative change is measured against the named comparator, and lower is better for MAE and WIS.

Stage	Data policy	Proposal	Comparator	Proposal	Change (%)	Decision
Point forecast	METR-LA test	Entangled VQC	STAEformer	3.8969	32.99	Fail
Point forecast	PEMS-BAY test	Entangled VQC	STAEformer	2.0295	31.84	Fail
QUARTS-UQ	METR-LA exploratory test	Entangled VQC	Separable VQC	2.0045	0.23	Fail
QUARTS-RAC	METR-LA validation	Entangled gate	Fourier gate	1.6986	0.70	Fail
RAQ	METR-LA validation	Asymmetric Fourier	Symmetric Fourier	1.7267	2.36	Fail
Prospective RAC	PeMSD4 test	Fourier gate	MLP gate	26.8820	-0.68	Provisional
Independent RAC	PeMSD4, three tests	Fourier gate	MLP gate	22.6892	-0.26	Fail
Recent calibration	PeMSD4 diagnostic	Recent Fourier	Ten percent window	21.9900	-1.88	Fail
SA-CQR	PeMSD8 validation	CQR-Fourier	CQR-MLP	15.2554	0.55	Fail

6.5 Calibration repair and PeMSD8 transfer

A recent window conformal diagnostic was evaluated for the failing seed 202 backbone. Increasing the recent calibration fraction from 10% to 30% reduced WIS from 22.4107 to 21.9900, but coverage reached only 88.09%. The repair therefore did not restore the coverage criterion.

The final transfer branch trained symmetric adaptive conformal quantile residual models on PeMSD8. Across three rolling validation folds, CQR-MLP obtained mean WIS 15.1716, while CQR-Fourier obtained 15.2554 and mean coverage 91.85%. Fourier was 0.55% worse and overcovered. According to the gate, the canonical PeMSD8 test split remained untouched. This result rejects RQ4 for the proposed transfer. Figure 7 distinguishes the post hoc repair from the protected validation only transfer.

6.6 Complete gate summary

Table 3 records the selected model, comparator, result, and decision for every stage. No entangled model passed a promotion gate. The prospective Fourier result was encouraging in one run but did not survive the stronger independent backbone criterion. The negative decisions are part of the result rather than discarded tuning attempts.

7 Discussion

7.1 What the controlled comparisons establish

The principal result is not that a quantum head failed to beat every classical model. It is that changing entanglement changed MAE by less than one quarter of one percent relative to the otherwise identical separable circuit, while both were far behind STAE-former. This pattern does not support an entanglement based explanation. The small improvement over Fourier is likewise insufficient because the unchanged backbone remained better and the effect size was negligible.

The result is consistent with the functional interpretation of data reuploading circuits. Periodic nonlinear features can be useful without being uniquely quantum [19]. A matched Fourier head exposes this alternative explanation directly. The architecture ablation further weakens a capacity argument: additional qubits and layers increased execution time but did not yield a consistent accuracy trend. The experiment cannot prove that entanglement is never useful for traffic data, but it bounds the claim for the tested local calibration setting.

7.2 Why the classical interval result did not qualify as final evidence

The first PeMSD4 run met its prospective WIS and coverage requirements and improved substantially over constant intervals. That finding demonstrates that the frozen latent state contains information about conditional residual scale. The margin over MLP was only 0.68%, however. Independent backbones exposed the fragility hidden by a single initialization. One seed reversed the ordering, and mean coverage fell below the accepted range.

This distinction matters for large traffic datasets. With more than one million paired forecasts per backbone, a very small mean difference can produce a narrow confidence interval. Statistical detectability does not guarantee stability across training conditions or practical relevance. The decision rule therefore combined paired uncertainty, effect direction, coverage, and independent initialization. Under that rule the Fourier model did not qualify as a robust improvement.

7.3 Implications for quantum transportation research

Three design principles follow. First, a quantum component should be assigned a clearly isolated role. A hybrid model that changes its backbone, input features, and optimization at the same time cannot identify the source of an improvement. Second, matched controls should reflect the known function class of the circuit. Parameter matching alone is not enough when a classical periodic layer captures the same spectral bias. Third, protected validation gates should remain active after a favourable result. The untouched PeMSD8 test set is more informative than a selectively reported transfer score because it shows that the stopping rule was respected.

Quantum traffic forecasting may still be valuable when a circuit processes a compact decision state, when data or kernels have a structure suited to a quantum feature

map, or when physical hardware changes the computational tradeoff. Those possibilities require hardware aware experiments, noise and shot budgets, strong network backbones, and repeated out of distribution evaluation. The present simulator results do not establish any of them.

8 Limitations and threats to validity

The quantum circuits were executed with analytic state vector simulation on a classical GPU environment. Consequently, runtime measurements describe the implementation used here and cannot predict latency on future quantum hardware. The primary circuits were also small. Larger circuits could represent different functions, but the observed runtime growth and nonmonotonic ablation make unrestricted scaling an unsupported remedy.

The point experiments use two speed networks, while the later interval studies use PeMSD4 and PeMSD8. This improves dataset diversity but does not cover incidents, weather disruptions, or long term distribution shift. Conformal calibration was applied chronologically and assessed empirically. Dependence between traffic observations means that standard exchangeability based coverage theory does not apply without qualification.

The paired error analysis contains many correlated node and horizon observations. Bootstrap intervals quantify the stored paired mean differences, but residual dependence can make them appear more precise than a replication level analysis. We therefore based the final PeMSD4 decision on three complete backbone initializations and directional consistency rather than on instance level significance alone.

STAEformer is a strong comparator on the selected datasets, but it is not the only modern forecasting architecture. Results could change with another frozen backbone or a different latent interface. Hyperparameter budgets were intentionally bounded and shared across treatments. A broader search might improve any head, yet it would also increase selection bias unless nested inside a new protected protocol.

Finally, the proposed architecture underwent a sequence of development stages. The record distinguishes confirmatory, exploratory, validation only, prospective, and post hoc analyses, but later hypotheses were informed by earlier outcomes. They should not be interpreted as one untouched confirmatory family. The complete gate table and archived configurations are supplied so readers can reconstruct this chronology.

9 Reproducibility and open science

The reproducibility package contains the complete source code, environment records, immutable experiment configurations, dataset integrity audit, seed level outputs, analysis scripts, publication tables, and vector figures. The package is available from the [QUARTS repository on GitHub](#). The repository commit associated with this manuscript gives commands for recreating every table and figure.

Raw benchmark data are not redistributed when their original terms or standard download routes apply. Instead, the package provides acquisition instructions, expected file layout, dimensions, and cryptographic hashes for the files used in the

experiments. Each run stores its dataset, split, seed, model treatment, metrics, timing, and completion state. Smoke tests and the preprotocol batch size 128 run are marked and excluded from publication aggregation.

The main analysis package expects 50 reference runs, 10 official STAEformer runs, and nine circuit ablations. Its completion audit reports no missing run. Validation only results are labelled separately from canonical test results. PeMSD8 test targets are absent by design because the validation gate failed. This absence can be verified from both the experiment ledger and the publication manifest.

10 Conclusion

QUARTS was designed to test a specific claim under controlled conditions: whether an entangled variational quantum circuit adds useful residual information to network traffic forecasts or predictive intervals. Across METR-LA and PEMS-BAY, it did not improve the unchanged backbone and remained about one third worse than STAEformer. Its differences from separable quantum and Fourier controls were below 0.40%, with negligible effect sizes. Circuit scaling increased cost without a consistent accuracy gain.

For uncertainty prediction, a Fourier regime calibrator achieved the strongest prospective result on one PeMSD4 backbone, reducing WIS by 14.58% relative to constant intervals and 0.68% relative to an MLP. The MLP comparison reversed for one of three independent backbones, and average 90% coverage fell to 88.78%. A recent conformal repair remained undercovered, and a conformal quantile Fourier model failed PeMSD8 validation. The protected test split was therefore not evaluated.

The study finds no empirical basis for attributing a forecasting or calibration advantage to entanglement in the tested system. Its positive result is methodological: strong backbones, functionally matched controls, seed level replication, interval aware metrics, and binding validation gates make a hybrid quantum claim more credible, whether the final evidence is favourable or negative.

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Declarations

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Competing interests. The authors declare that they have no competing interests.

Ethics approval. Not applicable. The study uses public traffic sensor datasets and does not involve human participants or animals.

Consent to participate. Not applicable.

Consent for publication. Not applicable.

Data availability. The datasets are publicly available from their established benchmark sources. File acquisition instructions and integrity hashes are included in the repository identified in Section 9.

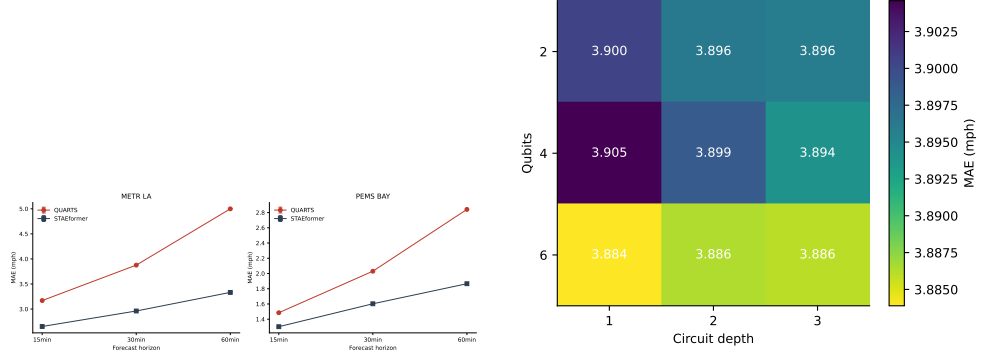


Fig. A1 Supplementary point results. Left: horizon specific error. Right: qubit and depth ablation.

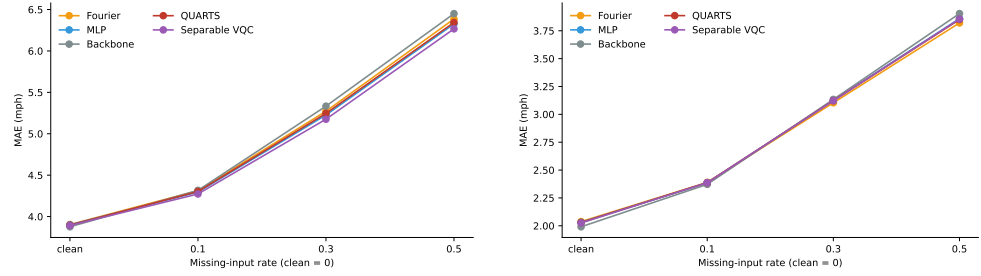


Fig. A2 Missing input sensitivity for METR-LA and PEMS-BAY.

Code availability. The complete code, configurations, tables, and figures are available from the repository identified in Section 9.

Author contributions. Author contribution roles will be added after author confirmation.

Appendix A Supplementary experiment record

The supplementary figures report horizon specific point errors, missing input robustness on METR-LA and PEMS-BAY, the quantum architecture ablation, simulated circuit noise sensitivity, the gated point pilot, the QUARTS-UQ and QUARTS-RAC studies, asymmetric interval validation, the initial PeMSD4 prospective run, additional PeMSD4 head seeds, and PeMSD8 SA-CQR validation. They are generated from the same publication manifest as the main figures.

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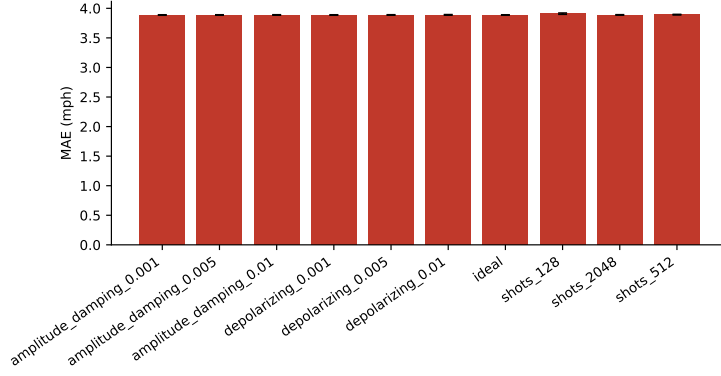


Fig. A3 Sensitivity of the simulated quantum head to the evaluated noise settings.

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